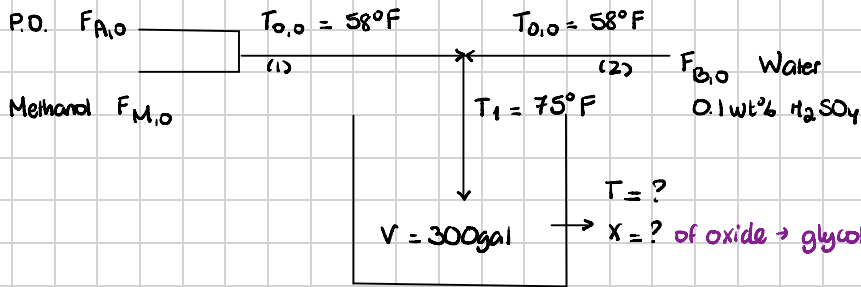


A = Propylene oxide, B = Water, C = Propylene glycol



$$\dot{m}_A = 2500 \text{ lb/h} \rightarrow F_A = 43.04 \text{ lb mol/h}$$



(1) Equimolar mixture of
46.62 ft³/h PO and 46.62 ft³/h MeOH

$$(2) \dot{V}_{H_2O} = 2.5 \dot{V}_{PO} = 233.1 \text{ ft}^3/\text{h}$$

$$\text{Thus, } F_{M,O} = 71.87 \text{ lb mol/h}$$

$$F_{B,O} = 802.8 \text{ lb mol/h}$$

The reaction is 1st order in A, 0 order in B.

$$K = A \exp\left(\frac{-E}{RT}\right) = 16.96 \times 10^2 \exp\left(\frac{-32,400}{RT}\right)$$

$$K [L = J h^{-1}], E [L = J \text{ lb mol}^{-1}]$$

Constraint to be neglected: Propylene oxide has low B.P. $\rightarrow$ DT of system $\leq 125^\circ\text{F}$. to avoid losing P.O. via vaporization

$$C_{p,A} = 35 \text{ Btu/lb mol}^\circ\text{F}; C_{p,B} = 18 \text{ Btu/lb mol}^\circ\text{F}; C_{p,C} = 46 \text{ Btu/lb mol}^\circ\text{F}; C_{p,M} = 19.5 \text{ Btu/lb mol}^\circ\text{F}$$

1) Mole balance and design equation

$$F_{A0} - F_A + r_A V = 0 \Rightarrow V = \frac{F_{A0} - F_A}{-r_A}$$

$$F_{A0} - F_A + F_{B0} - F_B + F_{M0} = 0$$

$$2) \text{ Rate law } -r_A = K C_A C_B$$

$$-r_A = K \underbrace{C_{A0}(1-X)}_{C_A} \underbrace{(C_{B0} - C_{A0} + C_{A0}X)}_{C_B}$$

3) Stoichiometry assuming liquid phase $v = v_0$

$$\text{Then } C_A = C_{A0}(1-X) \Rightarrow -r_A = K C_{A0}(1-X)$$

$$4) \text{ Combining: } V = \frac{F_{A0} X}{K C_{A0}(1-X)} = \frac{v_0 X}{K(1-X)} \Rightarrow \frac{V}{v_0} = \frac{X}{K(1-X)} \quad (1)$$

$$\text{Residence time: } \tau = \frac{V}{v_0} \quad (2)$$

$$(1) \text{ and } (2): \tau = \frac{X}{K(1-X)} \Rightarrow X = \frac{\tau K}{\tau K + 1} = \frac{\tau A \exp(-E/RT)}{1 + \tau A \exp(-E/RT)}$$

$$\begin{aligned} -r_A &= K C_{A0}(1-X)[C_{B0} - C_{A0} + C_{A0}(1-X)] \\ &= K C_{A0}(1-X)(C_{B0} - C_{A0}X) \\ &= K C_{A0}^2 (1-X) \left(\frac{C_{B0}}{C_{A0}} - X \right) \end{aligned}$$

5) Energy balance for adiabatic reaction with negligible energy input provided by the stirrer

$$X_{EB} = \frac{\sum \theta_i C_{p,i} (T - T_{i,0})}{-[\Delta H_{rxn}^\circ(T_R) + \Delta C_p (T - T_R)]} \quad (4)$$

$$\begin{aligned} &= (K C_{A0} - K C_{A0} X)(C_{B0} - C_{A0} X) \\ &= K C_{A0} C_{B0} - K C_{A0}^2 X - K C_{A0} C_{B0} X + K C_{A0}^2 X^2 \\ &= K C_{A0} X^2 + (-K C_{A0}^2 - K C_{A0} C_{B0}) X + K C_{A0} C_{B0} \end{aligned}$$

6) Calculations

a) Heat of reaction at T:

$$\Delta H_{rxn}(T) = \Delta H_{rxn}^\circ(T_R) + \Delta C_p (T - T_R)$$

$$H_A^\circ(68^\circ\text{F}) = -66600 \text{ Btu/lb mol}; H_B^\circ(68^\circ\text{F}) = -123000 \text{ Btu/lb mol}; H_C^\circ(68^\circ\text{F}) = -226000 \text{ Btu/lb mol}$$

$$\Rightarrow \Delta H_{rxn}^\circ(68^\circ\text{F}) = H_C^\circ - H_B^\circ - H_A^\circ = -36400 \text{ Btu/lb mol propylene oxide}$$

$$\Delta C_p = C_{p,C} - C_{p,B} - C_{p,A} = 46 - 18 - 35 = -7 \text{ Btu/lb mol}^\circ\text{F}$$

$$\text{Then: } \Delta H_{rxn}^\circ(T) = -36400 - 7(T - 528) \quad T \text{ is in } ^\circ\text{R}$$

b) Stoichiometry (C_{A0}, θ_i, τ)

The total liquid volumetric flowrate entering the reactor:

$$v_0 = v_{A0} + v_{M0} + v_{B0} = 46.62 + 46.62 + 233.1 = 326.3 \text{ ft}^3/\text{h}$$

$$V = 300 \text{ gal} = 40.1 \text{ ft}^3 \Rightarrow \tau = \frac{V}{v_0} = \frac{40.1 \text{ ft}^3}{326.3 \text{ ft}^3/\text{h}} = 0.123 \text{ h}$$

$$C_{A0} = \frac{F_{A0}}{v_0} = \frac{43.0 \text{ lb mol/h}}{326.3 \text{ ft}^3/\text{h}} = 0.132 \text{ lb mol/ft}^3$$

$$\text{For MeOH: } \theta_M = \frac{F_{M,0}}{F_{A,0}} = \frac{71.87 \text{ lb mol/h}}{43.04 \text{ lb mol/h}} = 1.67$$

$$\text{For water: } \theta_B = \frac{F_{B,0}}{F_{A,0}} = \frac{802.8 \text{ lb mol/h}}{43.04 \text{ lb mol/h}} = 18.65$$

$$\tau = \frac{X}{K X^2 + (-K C_A - K C_{B0}) X + K C_{B0}}$$

$$\begin{aligned} K \tau (X^2 - (C_{A0} + C_{B0}) X + C_{B0}) &= X \\ \Rightarrow K \tau X^2 - [K \tau (C_{A0} + C_{B0}) + 1] X + K \tau C_{B0} &= 0 \end{aligned}$$

c) Conversion from mole balance: from (3)

$$X_{MB} = \frac{(2.084 \times 10^{12}) \exp(-16306/T)}{1 + (2.084 \times 10^{12}) \exp(-16306/T)}, T \text{ is in } ^\circ R.$$

d) Evaluate balance terms

$$\sum \theta_i C_{p,i} = C_{pA} + \theta_B C_{pB} + \theta_C C_{pC} = 35 + (18.65)(18) + (1.67)(19.5) = 403.3 \text{ Btu/lbmol}^\circ F$$

$$T_o = T_{o,o} + \Delta T_{mix} = 58^\circ F + 17^\circ F = 75^\circ F = 535^\circ R$$

$$T_R = 68^\circ F = 528^\circ R.$$

$$\text{From (4): } X_{EB} = \frac{(403.3 \text{ Btu/lbmol}^\circ F)(T - 535)^\circ F}{-[-36400 - 7(T - 528)] \text{ Btu/lbmol}} = \frac{403.3(T - 535)}{36400 + 7(T - 528)}$$

Vary T , Find X_{MB} and X_{EB}
 ↓ ↓
 Material balance Energy balance

Plot X versus T ($^\circ R$).

$$-r_A = KCAC_B$$

$$K = K_o \exp\left(-\frac{E}{RT}\right)$$

$$C_A = C_{A0}(1-X)$$

$$K^* X^2 - [K^*(C_{A0} + C_{B0}) + 1]X + K^* C_{B0} = 0$$

$$X = \frac{aK + b}{cK + d}$$

$$\Rightarrow c \underline{KX} + dX = aK + b$$

$\Rightarrow$

Input: $T, C_{A0}, C_{B0}, C_{C0}$ Output: C_A, C_B, C_C

1st: Train NN where all constraints included, linearize where needed

↳ see the error

↳ interpret

2nd: Take out 1 eqn, linearized constraint.

$$F_{A0} - F_A + r_A V = 0$$

$$F_{B0} - F_B + r_B V = 0$$

$$F_{A0} + F_{B0} + F_{C0} = F_A + F_B + F_C$$



Stoichiometry: 1:1:1

1) Material Balance

$$\tau = V/Q$$

For A: $C_{A0}Q - C_AQ + r_A V = 0$
 $\Rightarrow C_{A0} - C_A + r_A \tau = 0 \quad (1)$

For B: $C_{B0} - C_B + r_B \tau = 0 \quad (2)$

Overall: $C_{A0} + C_{B0} = C_A + C_B + C_C \quad (3)$ (assuming no C initially)

$$\Rightarrow C_C = C_{A0} - C_A + C_{B0} - C_B$$

$$r_A = r_B = -r_C = -k_f C_A C_B + k_r C_C \quad (4)$$

$$k_f = A_{f,0} \exp\left(\frac{-E_{a,f}}{RT}\right) \quad (5) \quad k_r = A_{r,0} \exp\left(\frac{-E_{a,r}}{RT}\right) \quad (6)$$

Define constants:

$$C_{A0} = 1 \text{ mol/L}$$

$$C_{B0} = 2 \text{ mol/L}$$

$$V = 10 \text{ L} ; Q = 1 \text{ L/s} \Rightarrow \tau = V/Q = 10 \text{ s}$$

Parameters to change for more "aggressive" non-linearity:

$$A_{f,0} = 10^{12} ; E_{a,f} = 30000 \text{ J/mol}$$

$$A_{r,0} = 10^{10} ; E_{a,r} = 15000 \text{ J/mol}$$

Inputs: $T, C_{A0}, C_{B0}, C_{C0}$ } To NN.

Outputs: C_A, C_B, C_C